

Topic	What is Covered	Session Type
Quantum Fundamentals	QC basics- Introduction to Quantum Computing, Properties, Gates	Theory
Quantum Algorithms	Deutsch's Algorithm, Grover's Algorithm, Shor's Algorithm, Concept of Quantum Speed-up, Grover's Algorithm.	Theory
What is Quantum Machine Learning?	Motivation for QML, classical ML vs QML, where quantum computing fits into the ML pipeline, key terminology and landscape overview	Theory
Overview of QML	Types of QML approaches (quantum-enhanced ML, quantum-native models), classical-quantum hybrid paradigm, current research directions	Theory
Quantum Data Encoding – Part I	Need for data encoding, basis encoding, amplitude encoding, angle encoding — concepts and circuit representations	Theory
Quantum Data Encoding – Part II	Advanced encoding schemes (higher-order/feature-map encoding), encoding trade-offs (qubit count vs circuit depth), practical encoding examples	Theory
Hands-on: Data Encoding	Implementing basis, angle, and amplitude encoding on a quantum SDK/simulator using sample datasets	Hands-on
Parametric Quantum Circuits (PQC) – What is PQC	Structure of PQCs, tunable parameters/gates, ansatz design, role of PQCs in QML models	Theory
How to Benchmark PQC	Benchmarking metrics (expressibility, entangling capability, trainability), barren plateaus, comparing ansatz performance	Theory
Hands-on: Building and Benchmarking a PQC	Constructing a PQC on hardware/simulator, running benchmarking metrics, interpreting results	Hands-on
Variational Quantum Algorithms (VQA)	VQA framework, cost function design, optimization loop (classical optimizer + quantum circuit), examples (VQE, QAOA relevance to QML)	Theory
Hands-on: Variational Quantum Algorithm	Implementing a simple VQA workflow end-to-end on simulator/hardware	Hands-on
Other QML Models	Quantum Neural Networks (QNN), Quantum Kernel Methods, Quantum Support Vector Machines, Quantum Boltzmann Machines — brief survey	Theory
Hands-on: QML Model Implementation	Training a basic QNN or quantum kernel-based classifier on a sample dataset	Hands-on
Applications, Challenges and Limitations of QML	Real-world application areas (finance, chemistry, optimization), current limitations (noise, scalability, data loading bottleneck), open research challenges	Theory

